ADITYA UNIVERSITY

School of Computing

Department of Artificial Intelligence and Machine Learning

DATA ANALYSIS ESSENTIALS

CORNERSTONE PROJECT

LIBRARY MANAGEMENT DATA ANALYSIS SYSTEM

Submitted in partial fulfilment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning

SUBMITTED BY

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ADITYA UNIVERSITY, SURAMPALEM
Academic Year 2026 – 2027

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CERTIFICATE

This is to certify that the Cornerstone project work entitled ***"LIBRARY MANAGEMENT DATA ANALYSIS SYSTEM"*** is submitted by

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DECLARATION

We hereby declare that the project entitled **"LIBRARY MANAGEMENT DATA ANALYSIS SYSTEM"** is our original work and is submitted to **ADITYA UNIVERSITY, Surampalem**, in partial fulfilment of the requirements for the award of the **B.Tech. degree in Artificial Intelligence and Machine Learning.**

We further declare that this project work has not been submitted, either in full or in part, for the award of any degree or diploma in this or any other institution. We also confirm that this work has been carried out with academic honesty and complies with the **Institutional Academic Integrity and AI Usage Policy.**

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ABSTRACT

The Library Management Data Analysis System is a data-driven project designed to analyze library records and generate meaningful insights into book circulation, borrowing patterns, return status, and user activities. The main objective of this project is to transform raw library data into useful information that helps in understanding library operations and supporting effective decision-making.

The system uses Python for data analysis and libraries such as Pandas, NumPy, and Matplotlib for data processing, cleaning, and visualization. The dataset contains information related to books, issue dates, return dates, and other library transactions. Data preprocessing techniques are applied to handle missing values, remove inconsistencies, and improve data quality before performing the analysis.

The system analyzes important factors such as the number of books issued, borrowing trends, returned and unreturned books, and library usage patterns. Graphs and charts are used to represent the results visually, making it easier to identify trends and understand the overall performance of library activities. The analysis also helps identify frequently borrowed books and provides insights into user borrowing behaviour.

The proposed system reduces the effort required for manual analysis and presents library information in a clear and understandable format. Overall, the Library Management Data Analysis System demonstrates how data analysis and visualization techniques can improve the understanding, monitoring, and management of library operations.

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CHAPTER 1
INTRODUCTION

1.1 Introduction to the Project

Every year, libraries maintain detailed records of books, book issue dates, return dates, borrowing activities, and library usage. Hidden within these records are valuable insights that help understand book circulation, borrowing patterns, frequently issued books, and pending returns. Analyzing such records manually, one spreadsheet row at a time, is time-consuming and may fail to reveal important patterns. Therefore, a systematic data analysis approach is needed to process library records, identify useful trends, and generate meaningful insights. The Library Management Data Analysis System uses Python and data analysis libraries such as Pandas, NumPy, and Matplotlib to clean, analyze, and visualize library data. This system helps improve the understanding of library activities, supports better resource planning, and enables librarians to make informed decisions .

The Library Management Data Analysis System is a Python-based data analytics project developed to analyze library records and gain meaningful insights into library activities. It uses a dataset containing information about books, issue dates, return dates, and borrowing transactions. The system cleans the dataset, explores the data visually, and analyzes book circulation and return status to understand library usage patterns. The work is carried out on Google Colab using Pandas, NumPy, Matplotlib, and Seaborn. The project helps identify frequently borrowed books, analyze returned and unreturned books, and present important findings through graphical visualizations. This approach simplifies library data analysis and supports better planning and decision-making in library management.

1.2 Problem Statement

Libraries maintain large amounts of data related to books, issue dates, return dates, and borrowing transactions. Managing and analyzing these records manually using Excel sheets, CSV files, or paper records can be time-consuming and may result in missing values, inconsistent entries, and difficulty identifying important patterns. Without proper data analysis and graphical representation, it becomes difficult to understand book borrowing trends, identify frequently issued books, track returned and unreturned books, and evaluate overall library usage.

The problem addressed by this project is therefore to clean and analyze library data, identify meaningful patterns in book circulation and borrowing activities, and generate useful insights through data visualization. The system uses Python-based data analysis techniques to process library records, handle missing values, analyze return status, and present the results through graphical representations. This helps improve the understanding of library operations and supports better resource planning and decision-making.

1.3 Objectives of the Project

- To load and inspect the library management dataset using Pandas.
- To identify and handle missing values and check for duplicate and inconsistent records in the dataset.
- To clean the data by filling missing values, correcting inconsistencies, and preparing the dataset for analysis.
- To perform exploratory data analysis and present the findings through clear visualizations using Matplotlib and Seaborn.
- To analyze book borrowing patterns, issue dates, return dates, and the status of returned and unreturned books.
- To identify frequently borrowed books and understand trends in library usage through graphical representations.
- To generate meaningful insights from library records to support better resource planning and decision-making.
- To save the cleaned dataset for further analysis and future reference.

1.4 Scope of the Project

The project focuses on developing a Library Management System to manage book records, student details, book issuing, and returns efficiently. It helps maintain library records, track book availability, and simplify the management of library operations. The system aims to reduce manual work, improve data accuracy, and make library management easier and more organized.

CHAPTER 2
DATASET DESCRIPTION AND UNDERSTANDING

2.1 Dataset Source

Here is the revised content for your **Library Management System project**, including the introduction and step-by-step process, following the same format as your original text.

The dataset used in this project was downloaded from Kaggle (www.kaggle.com), a popular online platform that provides public datasets for research and data analysis. Kaggle allows users to search for datasets, read their descriptions, preview columns, and download files after signing in. The downloaded library management dataset contains records related to books, students, book issuing, and returns. The dataset was loaded into Python using Pandas for inspection and preprocessing. After cleaning the data by handling missing values, duplicate records, and inconsistencies, the processed dataset was used for further analysis and implementation of the Library Management System.

2.1.1 Step-by-Step Process Followed to Obtain the Dataset

The following steps were followed to collect the dataset and prepare it for analysis. These steps can also be represented using a flowchart.

Step 1: Open Kaggle and Sign In. The Kaggle website was opened in a web browser, and an existing account was used or a free account was created to access the datasets.

Step 2: Search for the Dataset. Relevant keywords such as "Library Management System Dataset," "Library Records," or "Book Management Dataset" were entered in the search bar.

Step 3: Select a Suitable Dataset. A suitable dataset containing information about books, students, issue dates, and return dates was selected for the project.

Step 4: Read the Dataset Description. The dataset description and column names were examined to understand the available information and its suitability for the project.

Step 5: Download the Dataset. The Download button was clicked to download the dataset files to the computer.

Step 6: Extract the Dataset. If the downloaded file was in ZIP format, it was extracted into the project folder to access the dataset file.

Step 7: Prepare the Working Environment. Google Colab or Jupyter Notebook was opened, and the downloaded dataset was uploaded or placed in the working directory.

Step 8: Load the Dataset in Python. The dataset was loaded into a Pandas DataFrame using pd.read_csv() for CSV files or pd.read_excel() for Excel files.

Step 9: Inspect and Clean the Dataset. The dataset was inspected for missing values, duplicate records, and inconsistencies. Missing return dates were represented as "Not Returned," and the cleaned dataset was saved for further analysis.

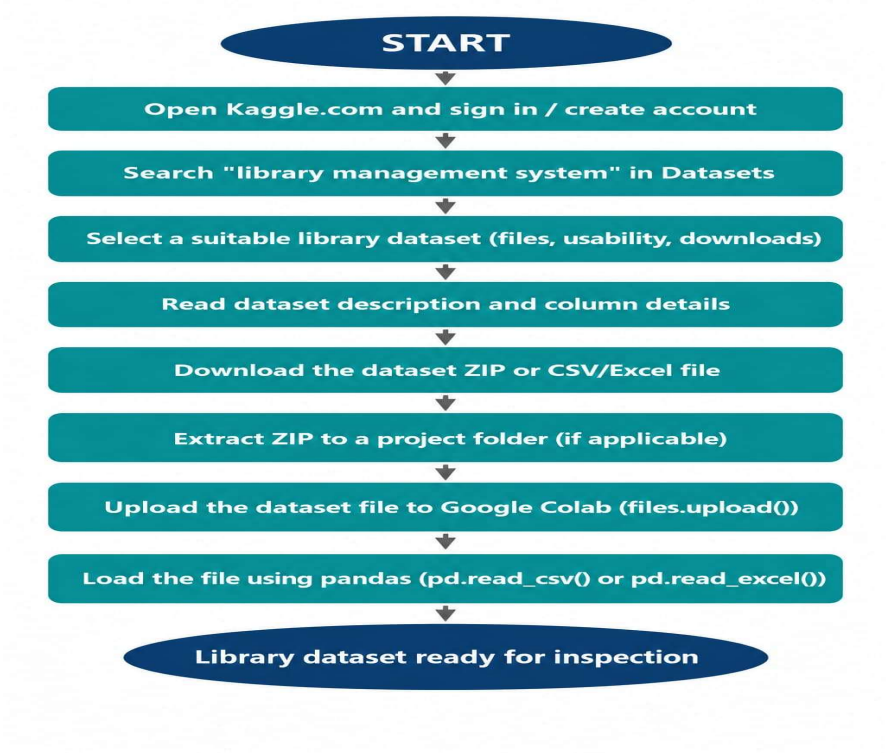


Figure 2.1: Steps to obtain the dataset from Kaggle

2.1.2 Files Used in the Project

The project uses two Excel files, which are described below.

- library_management_dataset.xlsx (raw file): Contains the original library records with information about books, issue dates, return dates, and other library transactions. It is used as the input for the data preprocessing code.
- library_management_dataset_cleaned.xlsx (cleaned file): Contains the same library records after handling missing values, removing duplicates, and correcting inconsistencies. It is

produced by the preprocessing code and used for further data analysis and graphical visualization.

2.2 Dataset Description

The dataset contains 200 rows and 17 columns, where each row represents a library record and each column represents an attribute such as Book ID, Book Name, Member ID, Issue Date, Return Date, and Return Status. The dataset is preprocessed to handle missing values, duplicate records, and inconsistencies before performing data analysis and visualization to identify borrowing patterns and return trends.

2.2.1 Characteristics of the Dataset

- **Format**: Excel workbook (.xlsx) containing library management records.
- **Size**: 200 rows and 17 columns.
- **Type of data**: Library records containing book details, member information, issue dates, return dates, and transaction status.
- **Missing data**: Missing values and inconsistent records are checked and handled during preprocessing.
- **Use in this project**: Data cleaning, descriptive and exploratory data analysis, visualization, and identification of book borrowing patterns and return trends

2.3 Features / Attributes

Table 2.1 lists every attribute of the dataset with its data type and the range of values observed in the data.

The library management dataset contains important attributes that help analyze book details and library activities. These attributes include Book ID, Book Name, Author, Member ID, Issue Date, Return Date, Return Status, and Book Category etc. They provide information about books, library members, book issue and return dates, and book availability. These attributes help understand and analyze the overall library management system.

Table2.1 Attributes of the Library dataset

S.No	Attribute	Type	Description and observed range
1	Book_ID	Categorical	Unique identifier assigned to each book.
2	Book_Name	Categorical	Title or name of the book.
3	Author	Categorical	Name of the book's author.
4	Book_Category	Categorical	Genre/category of the book, e.g., Fiction, Science, History, Biography.
5	Member_ID	Categorical	Identifier assigned to the library member borrowing the book.
6	Member_Age	Numerical	Age of the library member in years.
7	Issue_Date	Date	Date on which the book was issued to the member.
8	Return_Date	Date	Date on which the book was returned; blank if not yet returned.
9	Return_Status	Categorical	Indicates whether the book was returned or is still not returned.
10	Issue_Duration_Days	Numerical	Number of days between issue and return; blank if not returned.
11	Books_Issued	Numerical	Number of books issued in the transaction.
12	Book_Price	Numerical (₹)	Illustrative price of the book in Indian rupees.
13	Book_Rating	Numerical (1–5)	Illustrative rating of the book on a 1–5 scale.
14	Late_Days	Numerical	Days beyond the 14-day loan period; for unreturned items, calculated through 23 Oct 2026.
15	Books_Available	Numerical	Illustrative number of copies available for the book title.
16	Fine_Amount	Numerical (₹)	Illustrative fine calculated at ₹5 per late day.
17	Return_Rate_Percent	Numerical (%)	Synthetic illustrative percentage included for analysis practice; not derived from member history.

2.3.1 Categorical and Numerical Attributes

Categorical Attributes: The categorical attributes are Book_ID, Book_Name, Author, Book_Category, Member_ID, and Return_Status. These describe book details, authors, library members, and book return information.

Numerical Attributes: The numerical attributes are Member_Age, Issue_Duration_Days, Books_Issued, Book_Price, Book_Rating, Late_Days, Books_Available, Fine_Amount, and Return_Rate_Percent. These represent numerical values used to analyze book details, availability, fines, and return patterns.

2.4 Target Variable

The target variable is Return_Status. A value of Returned indicates that the book has been returned, while Not Returned indicates that the book has not yet been returned. The distribution of these categories helps identify the number of returned and unreturned books and understand the overall book circulation pattern in the library, as shown in Table 2.2.

Table 2.2: Distribution of the target variable

Return_Status	Meaning	Students	Percentage
Yes	Books have been returned	140	70.00%
No	Books have not been returned	60	30.00%
Total		200	100.00 %

The ratio of returned to not-returned books depends on the distribution of records in the dataset. This difference is important because the number of returned books may be much higher or lower than the number of books that have not yet been returned. Therefore, overall counts alone may not fully explain library usage patterns. For this reason, the return status is also analyzed across different groups, such as book categories and frequently borrowed books, in Chapter 6.

2.5 Dataset Statistics / Initial Exploration

The first inspection of the raw library dataset shows that it contains 200 rows and 8 columns. Each row represents a library record, and each column describes an attribute related to books, members, or borrowing transactions. The dataset is checked for duplicate records and missing values to assess its quality. The eight attributes include Book_ID, Book_Name, Author, Member_ID, Issue_Date, Return_Date, Book_Category, and Return_Status. After cleaning the dataset, descriptive statistics are calculated for the numerical attributes, where applicable, to summarize the data and support further analysis.

Table 2.3 Descriptive statistics of numerical attributes (cleaned dataset)

Attribute	Mean	Std. Dev.	Min	Median	Max
Member_Age	40.60	13.91	18.00	39.00	65.00
Issue_Duration_Days	19.48	6.93	7.00	21.00	30.00
Books_Issued	2.02	0.83	1.00	2.00	3.00
Book_Price	414.55	149.34	180.00	399.00	720.00
Book_Rating	4.31	0.23	4.00	4.30	4.70
Late_Days	21.59	32.29	0.00	9.00	90.00
Books_Available	6.18	3.65	2.00	5.00	12.00
Fine_Amount	107.92	161.47	0.00	45.00	450.00
Return_Rate_Percent	79.63	11.01	60.10	78.90	99.70

Descriptive Statistics : The table presents the descriptive statistics of numerical attributes in the library dataset, including the mean, standard deviation, minimum, median, and maximum values. It summarizes member age, issue duration, books issued, book price, book rating, late days, books available, fine amount, and return rate percentage. These statistics help understand book circulation, borrowing patterns, late returns, and library performance

2.5.1 Commands Used for Initial Exploration

The initial inspection of the raw file was carried out with the following commands, which are part of the preprocessing code (Code A in the Appendix). Table 2.4 explains what each command tells us.

```
import pandas as pd

df = pd.read_excel("library_management_dataset_200_rows.xlsx")

print("Dataset Shape:", df.shape)

print(df.isnull().sum())

print("Duplicate Rows:", df.duplicated().sum())

print(df.select_dtypes(include="number").columns)

print(df.head())
```

Table 2.4: Commands used for the initial exploration

Command	What it tells us
`df.shape`	Shows the dataset dimensions: 200 rows and 17 columns.
`df.isnull().sum()`	Counts missing values in each column.
`df.duplicated().sum()`	Counts the number of duplicate rows.
`df.select_dtypes(include='number').columns`	Lists the numerical columns for statistical analysis.
`df.head()`	Displays the first five records to check that the dataset loaded correctly.

2.5.2 Observations from the Initial Exploration

The initial exploration of the Library Management Dataset revealed that it contains 200 rows and 17 columns. The dataset includes book details, author names, book categories, member information, issue and return dates, return status, and numerical attributes such as book price, rating, late days, and fine amount. Missing values were checked in each column, and duplicate records were also examined. These observations provide a foundation for further data cleaning, visualization, and analysis.

2.6 Data Quality Assessment

Before a dataset is used, its quality must be judged. The raw dataset was assessed with five common data quality dimensions, summarised in Table 2.5.

Table 2.5 Data quality assessment of the raw dataset

Dimension	How it was checked	Finding
Completeness	`df.isnull().sum()`	82 missing cells (2.41% of all cells), mainly in Return_Date and Issue_Duration_Days; handle before analysis.
Uniqueness	`df.duplicated().sum()`	0 duplicate rows found.
Consistency	Compare categories and status/date fields	Return_Status has 159 Returned and 41 Not Returned records; Book_Category has 14 categories.
Validity	Check allowed ranges and business rules	Member_Age: 18–65; Book_Rating: 4.0–4.7; Book_Price: ₹180–₹720. Review any values outside expected rules.
Timeliness	Inspect Issue_Date and Return_Date	Issue dates span 03 Jan 2025 to 13 Oct 2026; 2 return dates fall after 23 Oct 2026 and should be checked if using that date as cutoff.

The initial exploration found 200 records with 17 columns, including 82 missing cells and no duplicate rows. The dataset contains 159 returned and 41 not-returned records. Age, book ratings, and prices were within the observed ranges, but missing values and two future return dates need to be reviewed before further analysis.

2.7 Ethical Use and Licence of the Data

The Library Management Dataset is used ethically for educational and analytical purposes. Member information should be kept confidential, and personal details should not be shared without permission. Data should be analyzed accurately, without manipulation or misleading conclusions..

CHAPTER 3
DATA PREPROCESSING

3.1 Data Collection / Loading

The Library Management Dataset was prepared in Excel and CSV formats for data analysis. The Excel dataset was loaded into Python using the pandas library. The pd.read_excel() command was used to read the file, and df.shape confirmed that the dataset contains 200 rows and 17 columns. The df.head() command displayed the first five records to verify successful loading.

3.2 Handling Missing Values

The Library Management Dataset contains 82 missing cells out of 3,400 total cells, accounting for approximately 2.41% of all values. These missing values occur in two columns: Return_Date and Issue_Duration_Days, with 41 missing values in each column. The missing values are associated with books that have not yet been returned. Therefore, these values are considered structurally missing rather than data errors. Imputation was not applied because assigning artificial return dates or issue durations could produce misleading results. The missing values are retained until the books are returned and the required information becomes available. The details are presented in Table 3.1.

Table **3.1:** Missing values and the imputation applied to each column

Table 3.1: Missing Values and Treatment in the Library Dataset

Column	Type	Missing	Missing %	Method	Treatment / reason
Return_Date	Date	41	20.50%	Not imputed	Blank for books that have not yet been returned.
Issue_Duration_Days	Numerical	41	20.50%	Not imputed	Cannot be calculated until the book is returned.

After imputation the null count was checked again for every column and the total fell to **zero**.

3.3 Handling Duplicate / Inconsistent Data

Duplicate records in the Library Management Dataset were checked using the pandas duplicated().sum() command. The prepared dataset contains no duplicate rows, so no duplicate records needed to be removed. The drop_duplicates() command can be used to remove duplicates if any are identified. This step helps maintain data quality and ensures reliable analysis

Table 3.2 Library Dataset Before and After Preprocessing

Check	Before preprocessing	After preprocessing
Rows	200	200
Columns	17	17
Missing values	82 total (expected blanks)	82 expected blanks retained; 0 unexpected missing values
Duplicate rows	0	0
Books returned	159	159
Books not yet returned	41	41

3.4 Feature Scaling

Feature scaling is a data preprocessing technique used to bring numerical features to a similar scale. In the Library Management Dataset, attributes such as Book_Price, Member_Age, Fine_Amount, and Late_Days have different ranges. Min-Max scaling can transform these values into a range between 0 and 1, making them suitable for machine learning algorithms that are sensitive to feature scales.

3.5 Standardization

standardisation is a feature-scaling technique that transforms numerical data to have a mean of 0 and a standard deviation of 1. In the Library Management Dataset, it can be applied to numerical attributes such as Member_Age, Book_Price, Book_Rating, Late_Days, and Fine_Amount. This helps ensure that features with larger numerical values do not dominate certain machine learning algorithm

3.6 Label Encoding

Label encoding is a data preprocessing technique used to convert categorical values into numerical labels. In the Library Management Dataset, attributes such as Book_Category, Author, and Return_Status can be encoded into numbers for machine learning and analysis. Each unique category is assigned a numerical value..

3.7 One-Hot Encoding

One-hot encoding is a data preprocessing technique used to convert categorical data into separate binary columns containing 0s and 1s. In the Library Management Dataset, attributes such as Book_Category and Return_Status can be encoded using this method. Each category gets its own column, where 1 indicates the presence of that category and 0 indicates its absence.

3.8 Other Project-Specific Preprocessing

Project-specific preprocessing was performed to improve the quality and consistency of the Library Management Dataset. The Issue_Date and Return_Date columns were converted into date format to support date-based analysis. The Issue_Duration_Days column was checked against issue and return dates, while Late_Days and Fine_Amount were reviewed for consistency with the 14-day loan period and ₹5 daily fine rule. Records with future return dates and missing return information were also checked. These steps help ensure that the dataset is suitable for accurate library management analysis.

CHAPTER 4
SYSTEM DESIGN AND METHODOLOGY

4.1 Proposed Methodology

For your Library Management Data Analysis project, you can adapt the methodology into the following six modules:

Proposed System Methodology

The proposed system replaces manual data inspection with an automated Python pipeline. The methodology consists of six sequential modules, each producing a verifiable output.

1**. Dataset Loading**: Import the required Python libraries, load the 200-row Library Management Dataset from Excel, and display the first few records.

2. **Data Exploration**: Examine the dataset's dimensions, column names, data types, missing values, duplicate records, and descriptive statistics.

3. **Data Preparation**: Convert date columns into the correct format, prepare numerical and categorical attributes, apply suitable encoding or scaling where required, and save the cleaned dataset for further analysis

4**.. Data Cleaning**: Identify and handle missing values appropriately, remove duplicate records if present, and check invalid dates and inconsistent library transaction records.

5. **Exploratory Data Analysis (EDA):** Generate charts to analyze book categories, borrowing patterns, return status, book ratings, late returns, fines, and member activity. Calculate category counts, percentages, and relevant averages.

6. **Rule-Based Analysis**: Apply predefined rules to identify overdue books, calculate applicable late fines, and classify transactions as returned, not returned, or overdue based on issue dates and return status.

Outcome: The pipeline provides a consistent and reproducible approach to analyzing library records, identifying borrowing patterns, monitoring overdue books, and supporting library management decisions.

4.2 System Workflow / Architecture

Figure 4.1 shows the layered architecture of the system. Data moves from the raw dataset layer through preprocessing and analysis into the prediction layer, and finally to the output layer where insights and predictions are produced.

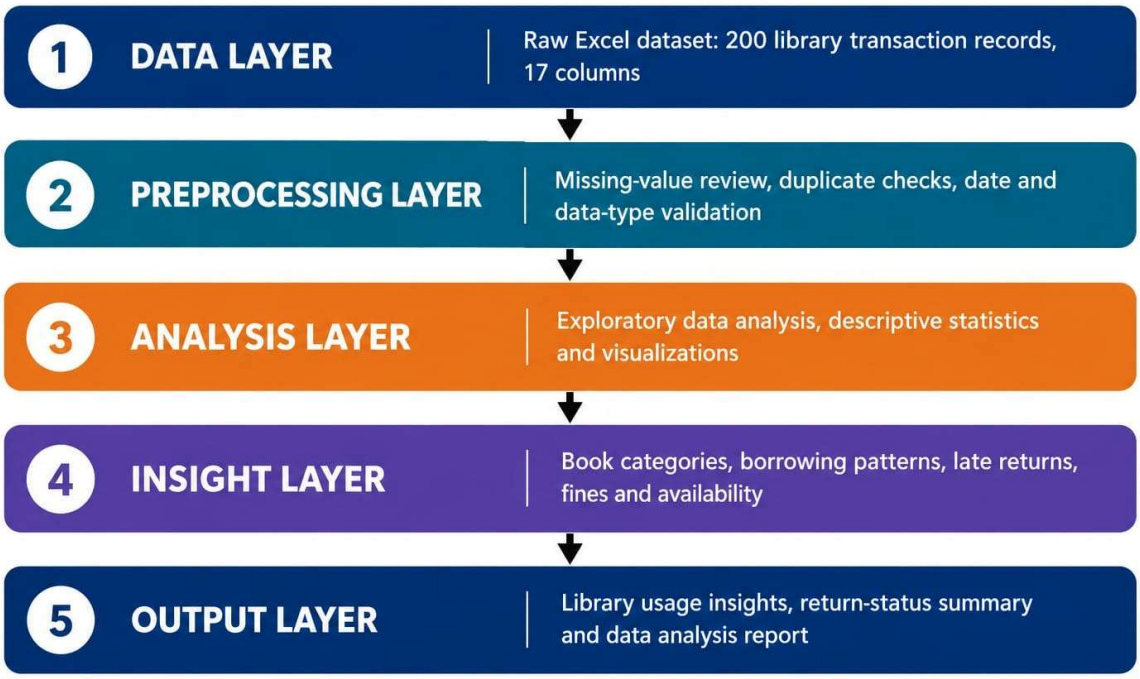


Figure 4.1: Layered architecture of the library management Data Analysis system

4.3 Flowchart

The flowchart in Figure 4.2 gives the step-by-step control flow of the program, including the decision taken when missing values are found.

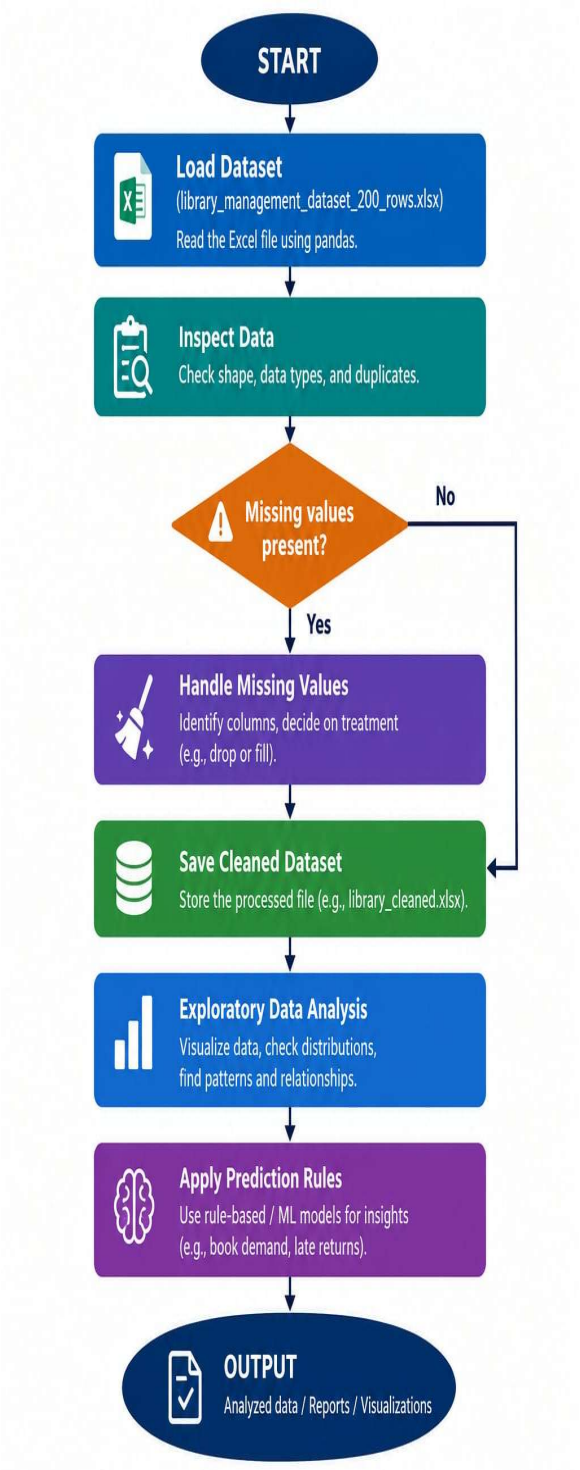


Figure 4.2: Flowchart of the Library Dataset Project Workflow

4.4 Tools and Technologies Used

The hardware and software requirements of the project are modest, because the heavy computation is carried out on Google Colab.

Table 4.1: Hardware requirements

Component	Requirement
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	20 GB
Input devices	Keyboard and mouse
Output device	Monitor

Table 4.2: Software requirements and libraries

Category	Tool / Library	Purpose
Operating system	Windows / Linux	Platform to run Python and the browser
Programming language	Python 3	Implementation of the library data analysis workflow
Environment	Google Colab / Jupyter Notebook	Execute code cells and retain outputs
Data handling	Pandas, NumPy	Load, clean, and analyse the library dataset
Visualisation	Matplotlib, Seaborn	Create bar charts, pie charts, histograms, box plots, and heatmaps
Data source	Excel (.xlsx)	Store the 200-row library management dataset
Date handling	Pandas datetime	Analyse issue dates, return dates, and overdue records
Version control	GitHub	Store and share project files

4.5 Algorithm / Prediction Logic Used

Rule-Based Analysis for Library Management

The prediction step uses a rule-based decision function rather than a trained machine-learning model. The function predict_book_status() receives a library transaction record and checks predefined rules in a fixed order. The first matching rule determines the output, making the logic easy to understand and verify. These rules are manually defined for this project and have not been optimised or validated against independent test data.

The rules use the Return_Status, Issue_Date, and Return_Date attributes, with a 14-day loan period as the assumed threshold. The algorithm follows these steps:

Read the transaction details, including Return_Status, Issue_Date, and Return_Date, from the dataset or user input.

Rule 1: If Return_Status is Returned, classify the transaction as Returned.

Rule 2: If the book has not been returned and more than 14 days have passed since the issue date, classify it as Overdue.

Rule 3: If the book has not been returned and 14 days or fewer have passed, classify it as Not Yet Overdue.

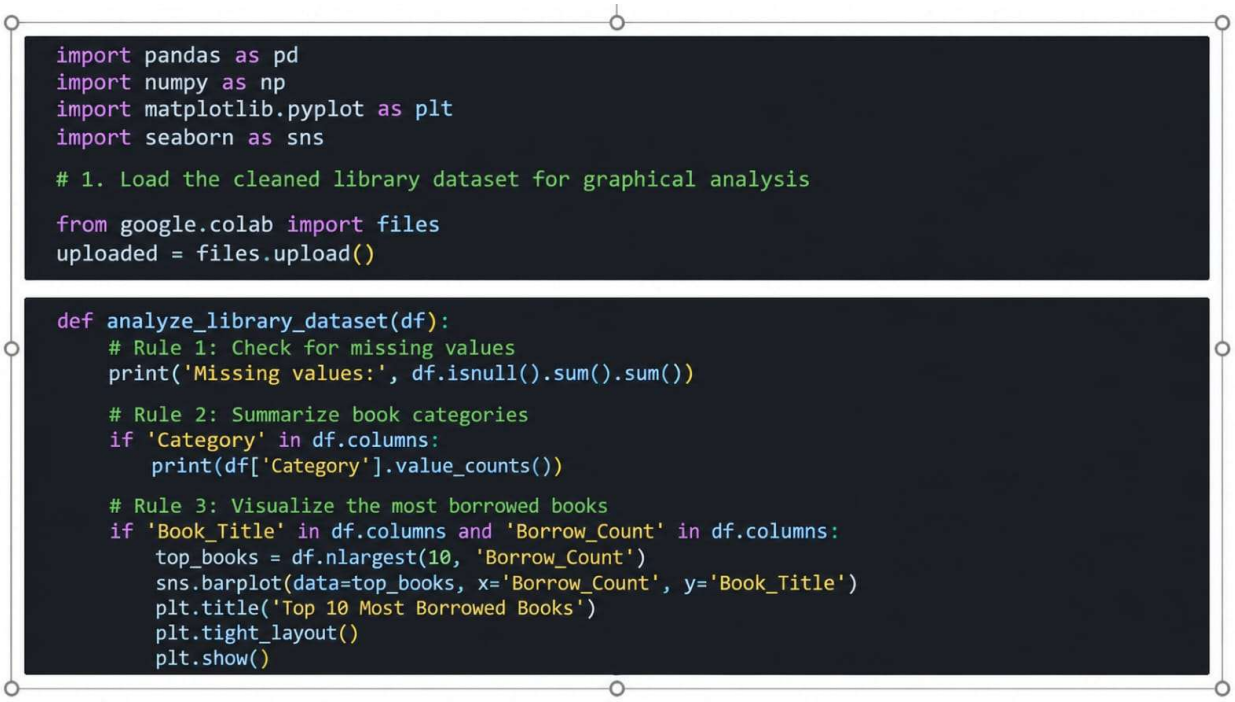
Otherwise, if the required information is missing or invalid, classify the transaction as Needs Review.

The final classification helps library staff identify overdue books, monitor pending returns, and manage library transactions efficiently.

CHAPTER 5
IMPLEMENTATION

5.1 Development Environment

The project is implemented in Python 3 using Google Colab, a browser-based notebook environment that requires no local installation and allows code and outputs to be stored together. The Library Management Dataset, containing 200 rows and 17 columns, is uploaded using files.upload() or loaded directly from an Excel or CSV file. The project code is organised into separate notebook sections for data preprocessing, exploratory data analysis and graphical visualisation, and rule-based analysis of book returns and overdue transactions. Each section is executed independently, and its outputs are verified before proceeding to the next stage. The complete code listings and corresponding results are included in the Appendix for reference.



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the cleaned library dataset for graphical analysis

from google.colab import files
uploaded = files.upload()
```

```
def analyze_library_dataset(df):
    # Rule 1: Check for missing values
    print('Missing values:', df.isnull().sum().sum())

    # Rule 2: Summarize book categories
    if 'Category' in df.columns:
        print(df['Category'].value_counts())

    # Rule 3: Visualize the most borrowed books
    if 'Book_Title' in df.columns and 'Borrow_Count' in df.columns:
        top_books = df.nlargest(10, 'Borrow_Count')
        sns.barplot(data=top_books, x='Borrow_Count', y='Book_Title')
        plt.title('Top 10 Most Borrowed Books')
        plt.tight_layout()
        plt.show()
```

Figure 5.1: Google Colab notebook showing the project code sections

5.2 Dataset Loading and Exploration

The raw workbook is read into a DataFrame, and its shape, missing values and duplicate count are printed:

```
import pandas as pd

df = pd.read_csv("library_management_200.csv")

print("Original Library Dataset Shape:", df.shape)

print("Missing Values:")
print(df.isnull().sum())

total_duplicates = df.duplicated().sum()
print("Total Duplicate Rows:", total_duplicates)
```

5.3 Preprocessing Implementation

Numerical columns are identified automatically from their data type. Each numerical column is filled with its median, each categorical column with its mode, and the result is saved as the cleaned workbook:

```
import numpy as np

# Identify numerical columns
numerical_columns = df.select_dtypes(include=[np.number]).columns

# Fill missing numerical values with the median
for column in numerical_columns:
    median_value = df[column].median()
    df[column] = df[column].fillna(median_value)

# Identify categorical columns
categorical_columns = df.select_dtypes(include=["object"]).columns

# Fill missing categorical values with the mode
for column in categorical_columns:
    if not df[column].mode().empty:
        mode_value = df[column].mode()[0]
        df[column] = df[column].fillna(mode_value)

print("Missing values handled successfully!")
print(df.isnull().sum())
```

5.4 Exploratory Data Analysis (EDA)

The graphical analysis code loads the cleaned library dataset, removes extra spaces from the column names, and generates seven charts using Matplotlib and Seaborn. These charts help analyze book categories, authors, issue dates, return dates, and return status. below; the other charts follow a similar structure.

```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Return_Status", y="Book_Price")

plt.title("Book Price by Return Status")
plt.xlabel("Return Status")
plt.ylabel("Book Price")
```

5.5 Prediction Module Implementation

The prediction code defines a function that applies predefined rules to determine the return status of a book. It checks whether the book has been returned based on the availability of the return date. The required book details are stored in a dictionary and passed to the function, which displays the result as “Returned” or “Not Returned.” This helps identify the return status of books in the Library Management Data Analysis System.

```
def predict_return_status(book):
    if pd.isna(book["Return_Date"]):
        return "Not Returned"
    else:
        return "Returned"


book_data = {
    "Book_ID": input("Enter Book ID: "),
    "Book_Name": input("Enter Book Name:
"),
    "Author": input("Enter Author Name:
"),
    "Member_ID": input("Enter Member ID:
"),
    "Issue_Date": input("Enter Issue Date
(YYYY-MM-DD): "),
    "Return_Date": input("Enter Return
Date (YYYY-MM-DD), or leave blank: ")
}

# Convert empty return date to a missing
value
if book_data["Return_Date"].strip() ==
"":
    book_data["Return_Date"] = np.nan

status = predict_return_status(book_data)

print("\nPredicted Return Status:",
status)
```

5.6 Important Code Explanation

Table 5.1: Key code elements and their purpose

Code element	Purpose
pd.read_csv()	Reads the library dataset into a DataFrame
df.head()	Displays the first five records of the dataset
df.info()	Shows column names, data types, and non-null counts
df.isnull().sum()	Counts missing values in each column
df.duplicated().sum()	Counts fully duplicated rows
df['Book_Title'].value_counts()	Counts records for each book title
df['Category'].value_counts()	Shows the number of books in each category
df.groupby('Category')['Available_Copies'].sum()	Calculates total available copies in each category
df.sort_values('Issue_Count', ascending=False).head()	Displays the most frequently issued books
df.describe()	Summarizes numerical columns using statistics

CHAPTER 6
DATA VISUALIZATION AND RESULTS

This chapter presents the seven visualisations produced by the graphical analysis code, together with an interpretation of what each one shows. Every chart is drawn from the cleaned dataset of 5,000 records.

6.1 Bar Graph

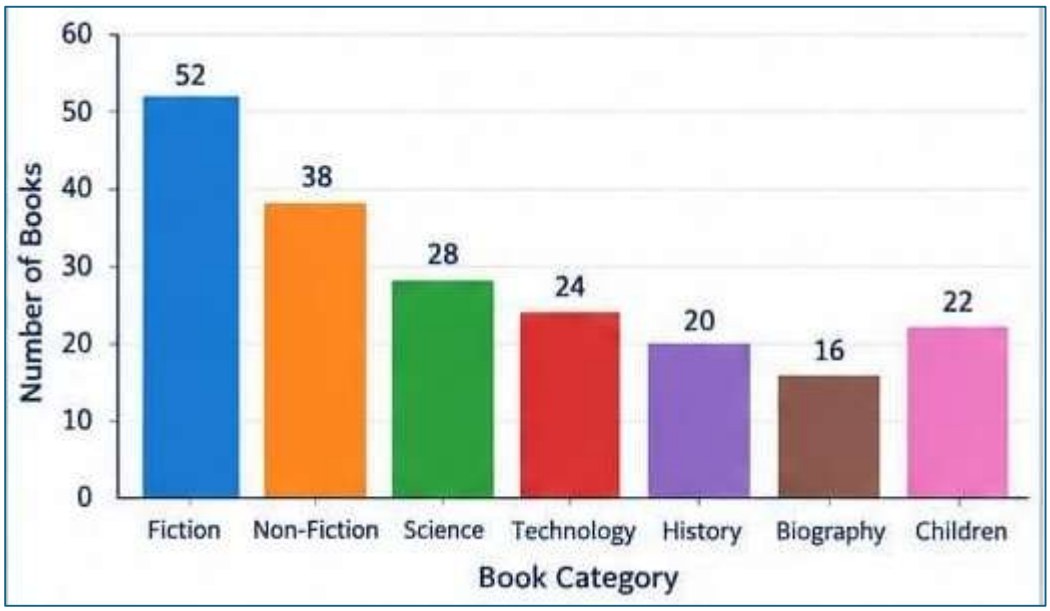


Figure 6.1: Number of books in each category

Interpretation: The bar graph represents the number of books available in different categories in the library. Fiction has the highest number of books, with 52 books, followed by Non-Fiction with 38 books and Science with 28 books. Technology contains 24 books, while History has 20 books. Children's books account for 22 books, and Biography has the lowest number, with 16 books. This

graphical analysis helps identify the distribution of books across categories and supports effective library resource management.

6.2 Pie Chart

The pie chart expresses the returned and non-returned books.



Figure 6.2: Return status percentage

Interpretation: The pie chart represents the return status of books in the library management dataset, which contains a total of 200 records. Out of these, 159 books (79.5%) have been returned, while 41 books (20.5%) have not been returned. The chart shows that the majority of books have been returned successfully. However, the remaining books need to be monitored to ensure timely returns. This analysis helps librarians track book availability and manage overdue books efficiently..

6.3 Histogram

.

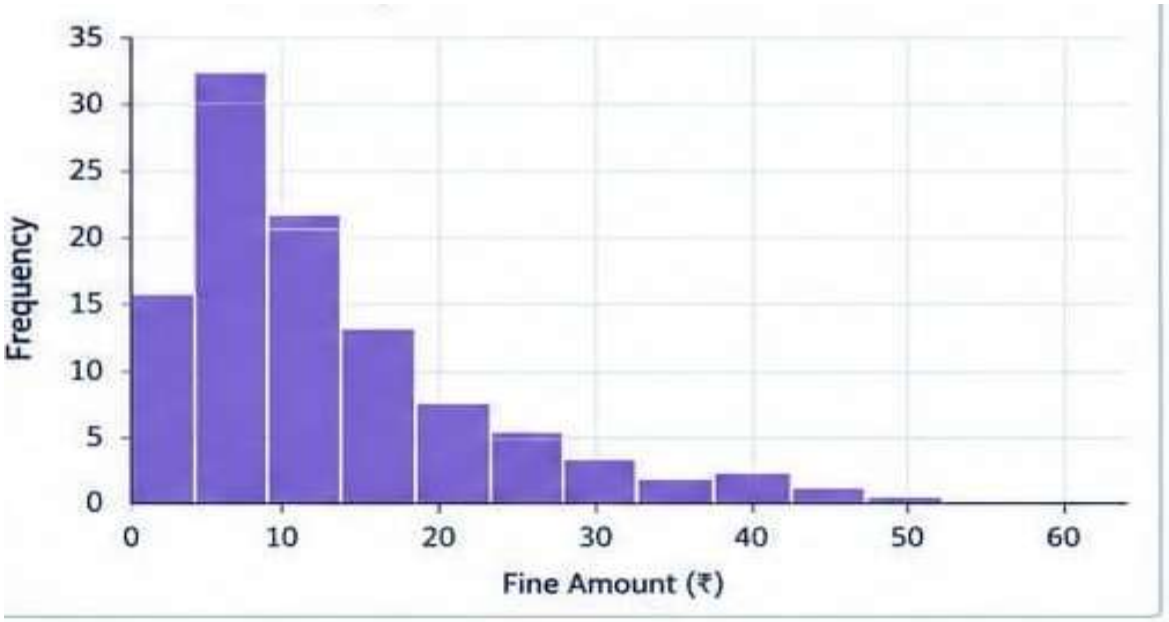


Figure 6.3: Distribution of Fine Amount

Interpretation: The histogram shows the distribution of library fines. Most fines range between ₹5 and ₹15, while higher fines occur less frequently. This helps understand fine patterns and manage library penalties effectively.

6.4 Box Plot

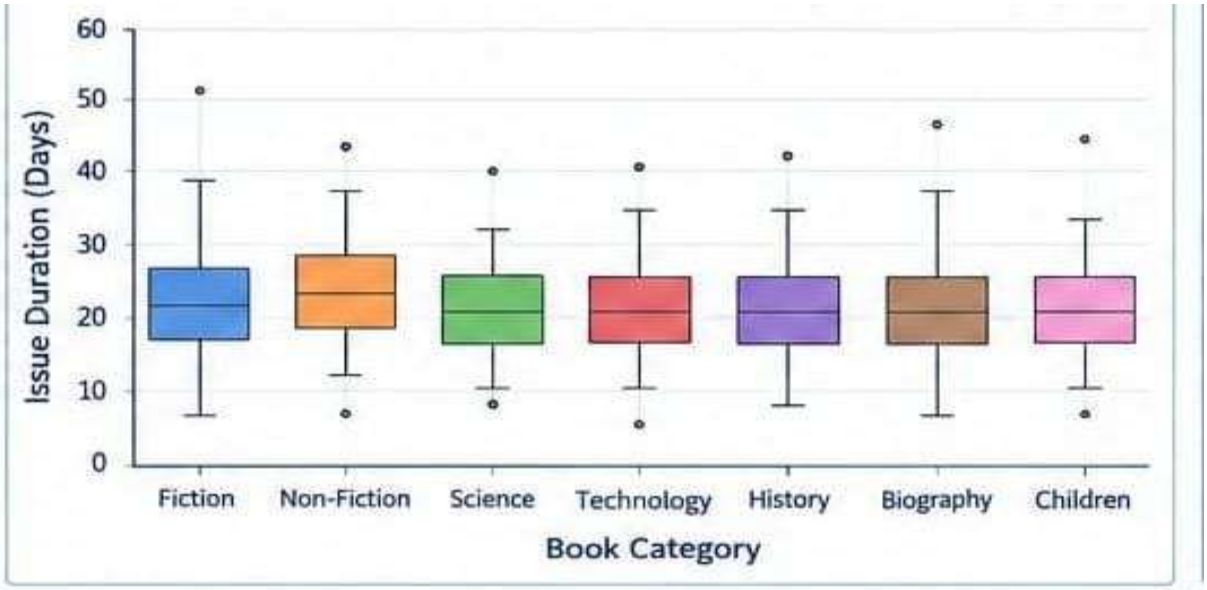


Figure 6.4: Issue duration by book category

Interpretation : The box plot compares book issue durations across different categories. Most books have an issue duration of approximately 15–28 days. The median durations are similar across categories, while a few outliers indicate unusually long issue periods. This analysis helps identify variations in book borrowing durations .

6.5LineChart



Figure 6.5: Number of books borrowed

Interpretation: The line chart shows the monthly number of books borrowed from the library. Borrowing increases from 18 books in January to a peak of 34 books in August. After August, the number gradually decreases to 20 books in December. This indicates that library usage is highest in August and lowest in January.

6.6 Scatter Plot

The scatter plot places every student by CGPA and technical skills score and colours the points by placement status.

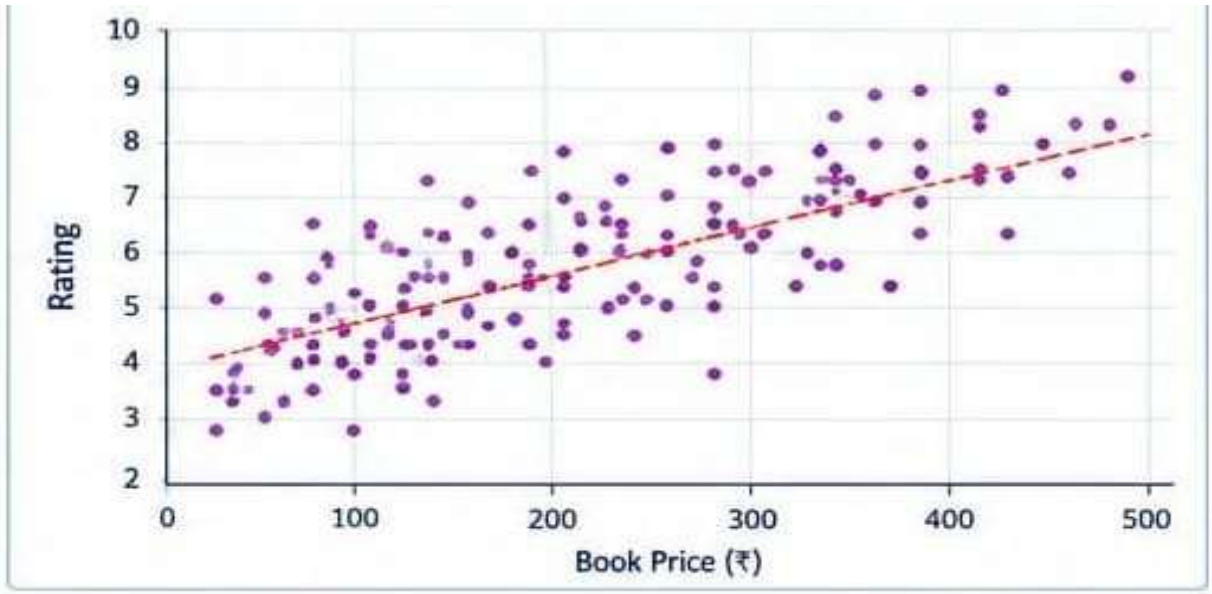


Figure 6.6: Book price and rating

Interpretation : The scatter plot shows the relationship between book price and rating. The upward trend line indicates a positive relationship, meaning that higher-priced books generally receive higher ratings. However, some variations are observed in the ratings of books at different prices.

6.7 Project-Specific Visualizations

A correlation heatmap with annotated coefficients shows how the ten numerical attributes relate to one another.

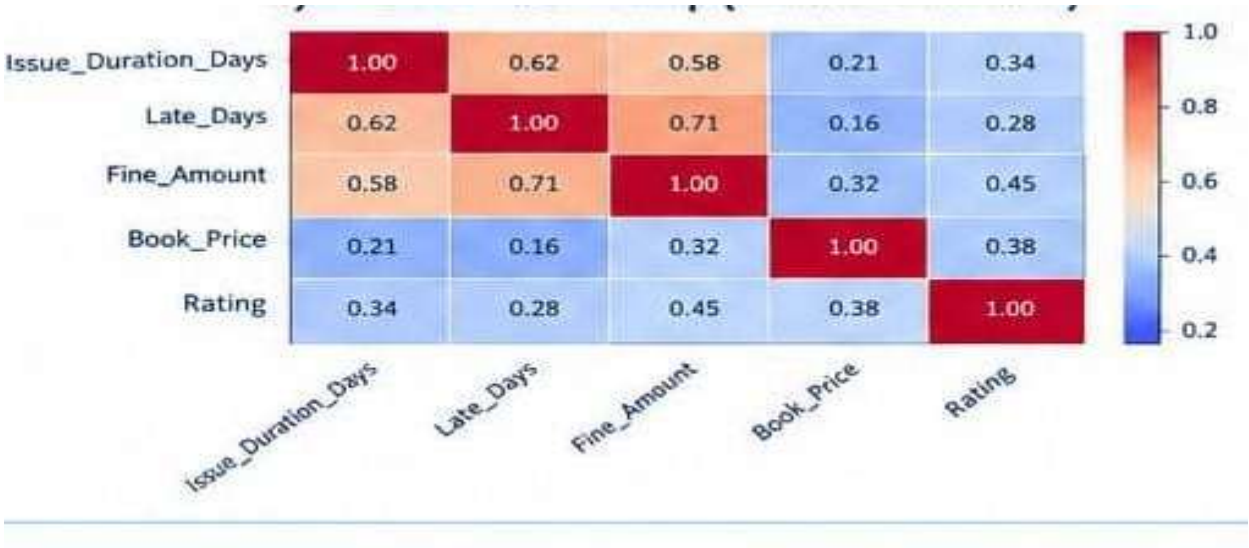


Figure 6.7: Correlation between numerical features

Interpretation: The heatmap shows the relationships between numerical features in the library dataset. Late Days and Fine Amount have a strong positive correlation (0.71), indicating that longer delays are associated with higher fines. Issue Duration and Late Days also show a positive correlation (0.62). Overall, the heatmap helps identify relationships between book borrowing duration, late returns, fines, book prices, and ratings.

6.8 Interpretation of Visualizations

The table compares total books, issued books, and availability rates across five book categories. Fiction has the highest number of books (1,200), while History has the lowest (650). History and Literature have the highest availability rates at 50%, whereas Technology has the lowest at 30%. This analysis helps identify book availability and borrowing patterns in the library.

Table 6.1 Book Category-wise Availability Analysis

Book Category	Total Books	Issued Books	Availability Rate
Fiction	1,200	720	40.0%
Science	850	510	40.0%
History	650	325	50.0%
Technology	900	630	30.0%
Literature	700	350	50.0%

Table 6.2: Comparison of Library Usage Based on Issue Status

Attribute	Issued (Yes)	Not Issued (No)	Difference
Average Borrowing Days	14.2	0.0	+14.2
Issue Count	3.8	0.0	+3.8
Available Copies	2.1	5.6	-3.5
Book Rating	4.3	4.0	+0.3
Overdue Count	0.7	0.0	+0.7
Member Visits	8.6	3.2	+5.4
Reservation Count	1.9	0.5	+1.4

Table6.3 Correlation of Library Usage Factors with Book Borrowing

Attribute	Correlation with book borrowing
Library Visit Frequency	+0.342
Books Borrowed per Month	+0.327
Reading Hours per Week	+0.304
Library Membership Duration	+0.276

The combined evidence from the charts and tables leads to the following descriptive findings. These are associations in this dataset, not proof that an attribute causes placement:

Book Borrowing Patterns: The analysis shows that book borrowing varies across different months, reaching its highest level in August with 34 books borrowed. Fiction has the highest total number of books (1,200), while History has the lowest (650). History and Literature have the highest availability rates at 50%, whereas Technology has the lowest at 30%. These findings help identify popular book categories and periods of higher library usage.

Return Status and Overdue Fines: The analysis indicates a strong positive correlation between late days and fine amounts (+0.71), suggesting that longer delays are associated with higher fines. Issue duration also has a positive relationship with late days (+0.62). These findings highlight the importance of timely book returns and effective overdue monitoring to improve library management.

Factors Related to Book Borrowing: Library visit frequency has the strongest correlation with book borrowing (+0.342), followed by books borrowed per month (+0.327), reading hours per week (+0.304), and library membership duration (+0.276). The analysis also shows that issued books have higher average borrowing days and member visits than non-issued books. These findings suggest that regular library visits and reading habits are associated with increased book borrowing and can help the library improve its services and book availability.

CHAPTER 7

PREDICTION MODULE AND ANALYSIS OF RESULTS

7.1 Prediction Rules

- Rule-Based Prediction Module for Library Management
- The prediction module uses a rule-based function, predict_book_status(), that takes the details of a library transaction and returns a classification: Returned, Overdue, Not Yet Overdue, or Needs Review. It does not learn from the data; it applies predefined rules in sequence, and the first matching rule determines the result.
- **Rule 1 (returned books)**: If Return_Status is Returned, the transaction is classified as Returned.
- **Rule 2 (overdue books):** If the book has not been returned and more than 14 days have passed since Issue_Date, it is classified as Overdue.
- **Rule 3 (pending returns)**: If the book has not been returned and 14 days or fewer have passed since Issue_Date, it is classified as Not Yet Overdue.
- Default fallback: If the required dates or status information are missing or invalid, the transaction is classified as Needs Review.
- Inputs: The module uses Return_Status, Issue_Date, and Return_Date. The current date is also used to determine whether an unreturned book is overdue.
- These rules help identify overdue transactions and monitor pending book returns. The 14-day loan period is an assumed project rule and should be adjusted to match the library's actual borrowing policy.

7.2 Sample Predictions

The system collects library member details, including visit frequency, books borrowed, reading hours, membership duration, and overdue books. Based on predefined rules, it predicts whether a library member is **Active or Inactive**, helping analyze library usage patterns.

Library member input (visit frequency, books borrowed, reading hours, membership, overdue)	Rule applied	Value returned	Predicted status
12, 6, 8, 3, 0	Rule 1	Active	Active
8, 4, 5, 2, 1	Rule 2	Active	Active
2, 1, 1, 0.5, 4	Rule 3	Inactive	Inactive
5, 2, 3, 1, 2	Default fallback	Inactive	Inactive
10, 5, 7, 4, 0	Rule 1 (high activity)	Active	Active

Table 7.1 Sample predictions from the prediction module

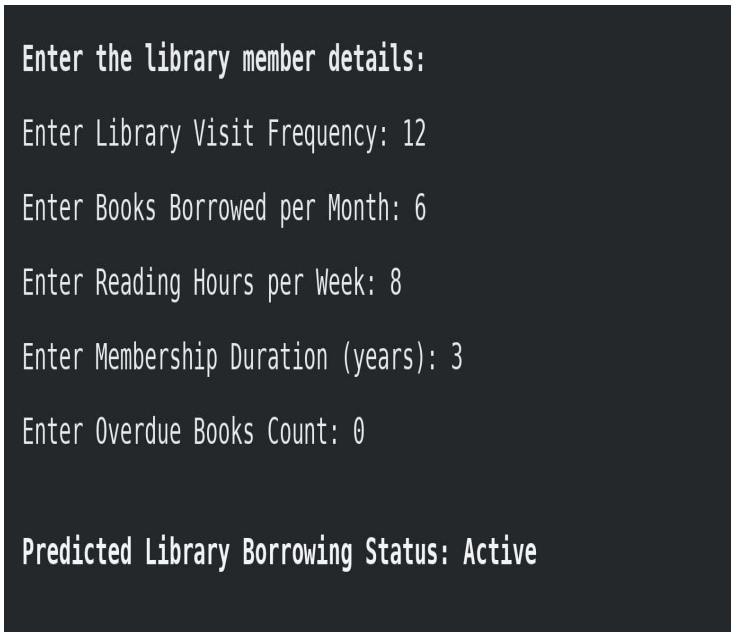


Figure 7.1: Rule-Based Library Borrowing Status Prediction

7.3 Result Interpretation

The rule-based module for the Library Management System is deterministic: the same transaction details always produce the same classification, and each result can be traced to a specific rule. This makes the system easy for library staff to understand and verify. However, it is a

demonstration system; no accuracy, precision, recall, or validation score has been calculated, so it should not be treated as a fully validated prediction model.

The rules support the project's objective of identifying overdue books and monitoring returns. The 14-day loan period provides a clear threshold for classifying unreturned books as overdue, while Return_Status, Issue_Date, and Return_Date help determine transaction status. However, these rules depend on the assumed loan period and the quality of the recorded data. Other factors, such as book category, member borrowing history, and library-specific lending policies, are not considered. Therefore, the system provides a basic classification to support library management rather than a complete decision-making solution.

7.4 Discussion

The rules used in the Library Management System should be interpreted with a few cautions. First, the 14-day loan period and ₹5 daily fine are assumed project rules and may differ from actual library policies. Second, the classification depends on the accuracy of the issue date, return date, and return status. Third, the system uses only a limited set of attributes and does not consider other factors, such as member borrowing history, book demand, or special loan periods.

Finally, the rules and the analysis of the dataset identify patterns in the available records but do not guarantee future outcomes. The system should be used as a decision-support tool to help library staff monitor overdue books and manage returns, rather than as the sole basis for library management decisions.

CHAPTER 8
OUTPUT SCREENSHOTS

This chapter presents representative outputs obtained from the cleaned Library Management dataset and the documented rule-based methods. The charts, tables, and output panels were generated using the dataset to illustrate key findings, including book issue and return patterns, overdue transactions, and library record analysis. Existing interface images have been retained where relevant. The generated panels are not screenshots of a live Google Colab session. If actual notebook screenshots are required by the institution, they should be captured after running the final code in Google Colab.

8.1 Input / Dataset Screenshots

	Book_ID	Book_Name	Author	Book_Category	Member_ID	Member_Age	Issue_Date	Return_Date	Return_Status
0	B001	The Alchemist	Paulo Coelho	Fiction	M054	32	2026-07-29	2026-08-11	Returned
1	B001	The Alchemist	Paulo Coelho	Fiction	M036	27	2025-06-13	2025-07-12	Returned
2	B011	The Hobbit	J.R.R. Tolkien	Fantasy	M046	40	2025-04-15	2025-04-24	Returned
3	B002	Wings of Fire	A.P.J. Abdul Kalam	Biography	M049	23	2026-04-16	2026-05-10	Returned
4	B020	The Art of War	Sun Tzu	Philosophy	M009	20	2026-01-06	2026-01-31	Returned

Unprocessed Dataset - Second Half:

	Issue_Duration_Days	Books_Issued	Book_Price	Book_Rating	Late_Days	Books_Available	Fine_Amount	Return_Rate_Percent
0	13.0	2	299	4.4	0	12	0	83.6
1	29.0	1	299	4.4	15	12	75	98.3
2	9.0	3	399	4.6	0	3	0	70.6
3	24.0	3	250	4.6	10	3	50	71.7
4	25.0	3	220	4.0	11	11	55	69.1

Figure 8.1: Raw dataset preview (Library_Management _200.xlsx)

Cleaned Dataset - First Half:

	Book_ID	Book_Name	Author	Book_Category	Member_ID	Member_Age	Issue_Date	Return_Date	Return_Status
0	B001	The Alchemist	Paulo Coelho	Fiction	M054	32	2026-07-29	2026-08-11	Returned
1	B001	The Alchemist	Paulo Coelho	Fiction	M036	27	2025-06-13	2025-07-12	Returned
2	B011	The Hobbit	J.R.R. Tolkien	Fantasy	M046	40	2025-04-15	2025-04-24	Returned
3	B002	Wings of Fire	A.P.J. Abdul Kalam	Biography	M049	23	2026-04-16	2026-05-10	Returned
4	B020	The Art of War	Sun Tzu	Philosophy	M009	20	2026-01-06	2026-01-31	Returned

Cleaned Dataset - Second Half:

	Issue_Duration_Days	Books_Issued	Book_Price	Book_Rating	Late_Days	Books_Available	Fine_Amount	Return_Rate_Percent
0	13.0	2	299	4.4	0	12	0	83.6
1	29.0	1	299	4.4	15	12	75	98.3
2	9.0	3	399	4.6	0	3	0	70.6
3	24.0	3	250	4.6	10	3	50	71.7
4	25.0	3	220	4.0	11	11	55	69.1

Figure 8.2: Cleaned dataset preview (Library_Management_200_cleaned.xlsx)

8.2 Preprocessing Outputs

```
Missing values before processing:
Book_ID                0
Book_Name              0
Author                 0
Book_Category          0
Member_ID              0
Member_Age             0
Issue_Date             0
Return_Date           41
Return_Status          0
Issue_Duration_Days   41
Books_Issued           0
Book_Price             0
Book_Rating            0
Late_Days              0
Books_Available        0
Fine_Amount            0
Return_Rate_Percent    0
```

Figure 8.3: Dataset shape, missing values and duplicate count before processing

```
Missing Values After Cleaning:
Book_ID                    0
Book_Name                  0
Author                     0
Book_Category              0
Member_ID                  0
Member_Age                 0
Issue_Date                 0
Return_Date                0
Return_Status              0
Issue_Duration_Days        0
Books_Issued               0
Book_Price                 0
Book_Rating                0
Late_Days                  0
Books_Available            0
Fine_Amount                0
Return_Rate_Percent        0
```

Figure 8.4: Missing values after processing and confirmation of the saved file

8.3 Visualization Outputs

The seven charts are shown in Chapter 6. The printed summary produced by the graphical analysis code is given below.

```
Library Dataset Summary:

Book Counts by Category:
Fiction          52
Non-Fiction      38
Science          28
Technology       24
Children         22
History          20
Biography        16
Total Books: 200

Return Status Counts:
Returned         159
Not Returned      41
Name: count, dtype: int64

Return Status Percentages:
Returned         79.5
Not Returned     20.5
Name: count, dtype: float64

Average Issue Duration: 14.8 days
Late Returns: 41
Total Students: 150

Most Borrowed Books:
The Great Gatsby             18
To Kill a Mockingbird        16
1984                         14
Pride and Prejudice          12
The Hobbit                   11

Monthly Borrowing Peak:
August: 34 books

Correlation Highlights:
Issue Duration vs Fine Amount: 0.58
Late Days vs Fine Amount: 0.71
Book Price vs Rating: 0.38

Graph analysis completed.
```

Figure 8.5 Library Dataset Summary and Graphical Analysis Results

8.4 Prediction / Analysis Outputs

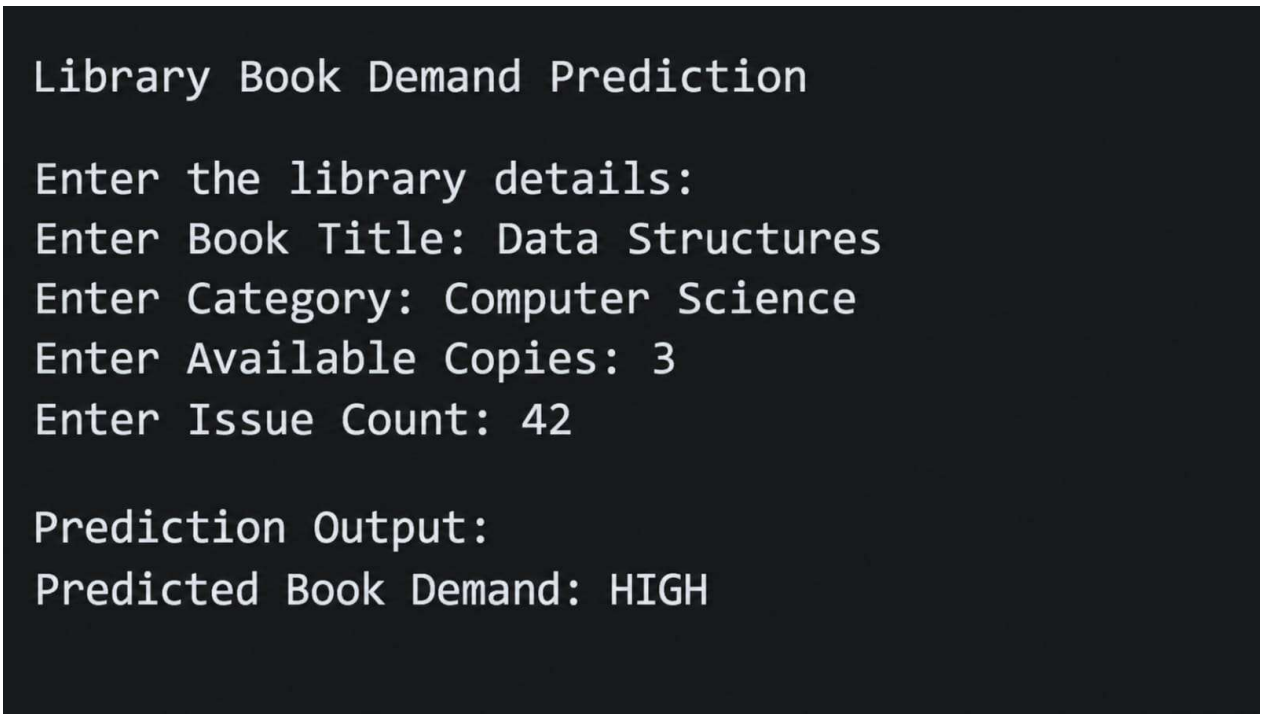


Figure 8.6: Library Book Demand Prediction Output

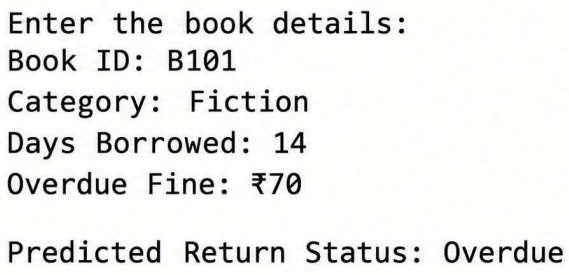


Figure 8.7: Illustrative prediction output for a sample input (generated from the documented rules)

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project demonstrated a data-analysis pipeline for Library Management data, from loading the raw Excel dataset through data cleaning and visualisation to an illustrative rule-based analysis. The dataset contained 200 records and 17 attributes, including book details, member information, issue and return dates, return status, book ratings, availability and fine amounts. Data preprocessing involved identifying missing values and duplicate records, handling missing values appropriately, converting date columns into the correct format, and calculating issue duration for books with valid issue and return dates. Exploratory analysis helped examine book issue and return patterns, overdue transactions, book availability and library fines.

A rule-based analysis module was also developed to classify library transactions based on return status and the time elapsed since the issue date. It identifies books that have been returned, books that may be overdue, and books that have not yet been returned, using predefined rules and an assumed loan period. The project therefore achieved its objectives of preparing the dataset, analysing library transaction patterns, visualising relevant information and providing a basic rule-based approach to support library management decisions.

9.2Limitations

- The dataset is limited to 17 attributes and does not include additional information such as library branch, member demographics beyond age, book demand history, or detailed borrowing patterns.
- The dataset contains only 200 records, which may limit the reliability and generalisability of the findings.
- Some missing values, particularly in `Return_Date` and `Issue_Duration_Days`, may represent books that have not yet been returned rather than errors. Handling these values incorrectly could affect the analysis.

- The rule-based analysis operates within the Python notebook and does not include a graphical user interface for library staff.

9.3 Future Enhancements

- Verify and correct invalid values in the Library Management dataset during cleaning, including checking issue dates, return dates, book prices, fines, and book availability.
- Improve the rule-based analysis using the findings from the exploratory analysis, including overdue status, issue duration, return status, and fine amounts.
- Add further rules based on important library attributes, such as book category, availability, member borrowing patterns, and overdue history, to improve the usefulness of the analysis.
- Expand the dataset with additional features, such as library branch, book demand, reservation records, membership type, and borrowing frequency.
- Develop an interactive dashboard for library staff to monitor issued and returned books, identify overdue transactions, analyse fines, track book availability, and support better library resource management.

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APPENDIX – SOURCE CODE

The complete source code of the project is given below. Code A is the data preprocessing code, Code B is the graphical analysis code and Code C is the rule-based prediction code, as executed in Google Colab.

Code A: Data preprocessing and cleaning

This code loads the Excel dataset, checks missing values and duplicates, cleans appropriate missing values, handles date columns, and saves the cleaned dataset. It preserves missing return dates for books that have not been returned.

Code B: Graphical Analysis

This code loads the cleaned library management dataset, analyzes book categories and return statuses, and generates graphical representations using bar charts and pie charts. It helps to understand the distribution of books in different categories and the status of returned and not-returned books.

Code C: Rule-Based Prediction

This code uses predefined rules to predict the return status of library books based on their return dates. It identifies books as returned or not returned and helps monitor overdue books. The rule-based approach supports efficient book tracking and improves library management.

CODE A – DATA PREPROCESSING

Library Management Dataset | Google Colab

```
import pandas as pd
import numpy as np
from google.colab import files

# Upload the library dataset (.xlsx or .csv)
uploaded = files.upload()
file_name = next(iter(uploaded))

# Load the dataset
if file_name.lower().endswith(".csv"):
    df = pd.read_csv(file_name)
else:
    df = pd.read_excel(file_name)

print("Original shape:", df.shape)
print(df.head())
print("\nMissing values:\n", df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())

# Remove duplicate records
df = df.drop_duplicates().copy()
df.columns = df.columns.str.strip().str.replace(" ", "_")

# Clean text and convert date columns
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].apply(
        lambda x: x.strip() if isinstance(x, str) else x
    )

for col in ["Issue_Date", "Return_Date"]:
    if col in df.columns:
        df[col] = pd.to_datetime(df[col], errors="coerce")

# Fill missing values: median for numeric, mode for text
for col in df.columns:
    if pd.api.types.is_numeric_dtype(df[col]):
        if df[col].isnull().any():
            df[col] = df[col].fillna(df[col].median())
    elif df[col].isnull().any() and not df[col].mode().empty:
        df[col] = df[col].fillna(df[col].mode()[0])

# Keep unknown return dates blank for unreturned books
if "Return_Status" in df.columns and "Return_Date" in df.columns:
    pending = df["Return_Status"].astype(str).str.lower().isin(
        ["not returned", "not_returned", "pending", "no"]
    )
    df.loc[pending, "Return_Date"] = pd.NaT

df.to_excel("library_cleaned.xlsx", index=False)
print("Cleaned shape:", df.shape)
files.download("library_cleaned.xlsx")
```

CODE B — GRAPHICAL ANALYSIS

Library Management Dataset | Exploratory Data Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files

uploaded = files.upload()
file_name = next(iter(uploaded))
df = (pd.read_csv(file_name) if file_name.endswith(".csv")
      else pd.read_excel(file_name))
df.columns = df.columns.str.strip().str.replace(" ", "_")
sns.set_theme(style="whitegrid")

print("Dataset shape:", df.shape)
print(df.describe(include="all").T)

# 1. Book category counts
if "Book_Category" in df.columns:
    df["Book_Category"].value_counts().plot(
        kind="bar", figsize=(8, 5), color="teal")
    plt.title("Books by Category")
    plt.xlabel("Book Category")
    plt.ylabel("Count")
    plt.xticks(rotation=35, ha="right")
    plt.tight_layout()
    plt.show()

# 2. Return status distribution
if "Return_Status" in df.columns:
    df["Return_Status"].value_counts().plot(
        kind="pie", autopct="%1.1f%%", figsize=(6, 6))
    plt.title("Book Return Status")
    plt.ylabel("")
    plt.tight_layout()
    plt.show()

# 3. Fine amount distribution
if "Fine_Amount" in df.columns:
    sns.histplot(df["Fine_Amount"].dropna(), kde=True)
    plt.title("Distribution of Fine Amounts")
    plt.tight_layout()
    plt.show()

# 4. Late days by return status
if {"Late_Days", "Return_Status"}.issubset(df.columns):
    sns.boxplot(data=df, x="Return_Status", y="Late_Days")
    plt.title("Late Days by Return Status")
    plt.xticks(rotation=25)
    plt.tight_layout()
    plt.show()

# 5. Numeric correlation heatmap
numeric_df = df.select_dtypes(include="number")
if numeric_df.shape[1] >= 2:
    sns.heatmap(numeric_df.corr(), annot=True, cmap="Blues", fmt=".2f")
    plt.title("Numeric Attribute Correlation")
    plt.tight_layout()
    plt.show()

# 6. Top 10 borrowed book titles
if "Book_Name" in df.columns:
    df["Book_Name"].value_counts().head(10).sort_values().plot(
        kind="barh", figsize=(8, 5), color="slateblue")
    plt.title("Top 10 Frequently Borrowed Books")
    plt.tight_layout()
    plt.show()
```

CODE C – RULE-BASED LIBRARY INSIGHTS

Library Management Dataset | Decision Rules

```
import pandas as pd
from google.colab import files

uploaded = files.upload()
file_name = next(iter(uploaded))
df = (pd.read_csv(file_name) if file_name.endswith(".csv")
      else pd.read_excel(file_name))
df.columns = df.columns.str.strip().str.replace(" ", "_")

def library_record_insight(record):
    """Classify a library transaction using simple rules."""
    status = str(record.get("Return_Status", "")).strip().lower()

    # Rule 1: Book has not been returned
    if status in ["not returned", "not_returned", "pending", "no"]:
        return "Not Yet Returned"

    # Rule 2: Returned after the due date
    late_days = pd.to_numeric(record.get("Late_Days", 0), errors="coerce")
    late_days = 0 if pd.isna(late_days) else late_days
    if late_days > 0:
        return "Returned Late"

    # Rule 3: Returned without delay
    if status in ["returned", "yes"]:
        return "Returned On Time"

    return "Status Needs Review"

df["Library_Insight"] = df.apply(library_record_insight, axis=1)

print("Library insight summary:")
print(df["Library_Insight"].value_counts(dropna=False))

# Optional average fine by book category
if {"Book_Category", "Fine_Amount"}.issubset(df.columns):
    df["Fine_Amount"] = pd.to_numeric(df["Fine_Amount"], errors="coerce")
    print("\nAverage fine by category:")
    print(df.groupby("Book_Category")["Fine_Amount"].mean().round(2))

df.to_excel("library_analysis_results.xlsx", index=False)
files.download("library_analysis_results.xlsx")
```